

5-Day Python Training Plan

Day 1 – Introduction & Python Basics

- Introduction to Python and Applications
- Installation & Setup (Python, VS Code, Jupyter)
- Python Syntax & Indentation
- Variables & Data Types (int, float, string, bool)
- Operators (Arithmetic, Comparison, Logical)
- Hands-on: Basic calculator, user input program, simple interest calculation

Day 2 – Control Flow & Functions

- Conditional Statements (if, elif, else)
- Loops (for, while, break, continue)
- Functions (parameters, return, default arguments)
- Lambda functions (introduction)
- Modules (math, random, custom modules)
- Hands-on: Prime number check, multiplication table, calculator module

Day 3 – Data Structures & File Handling

- Lists (methods, slicing, list comprehension)
- Tuples & Sets (properties, operations)
- Dictionaries (creation, looping, methods)
- File Handling (read, write, append, with statement)
- Working with CSV (introduction)
- Hands-on: Student dictionary, file read/write, word count program

Day 4 – Exception Handling & OOP

- Exception Handling (try, except, finally, raise)
- Classes & Objects
- `__init__` method and instance variables
- Inheritance & Method Overriding
- Encapsulation & Polymorphism (basics)
- Hands-on: Student class, Employee inheritance program, exception handling example

Day 5 – NumPy, Pandas & Mini Project

- NumPy Basics (arrays, operations, mean, sum)
- Pandas Basics (Series, DataFrame)
- Reading & Filtering CSV Data
- Basic Data Analysis
- Mini Project: Student Management System OR Sales Data Analysis
- Hands-on: Data analysis using NumPy & Pandas